✅ PHASE-WISE CHECKLISTS (2-Week Blocks)

PHASE 1 (Weeks 1–2) — Hardware Planning & Ordering

Checklist:

- [ ] Create preliminary BOM (Bill of Materials)
- [ ] Confirm microcontroller choice (e.g., Teensy 4.0)
- [ ] Confirm motor/driver compatibility
- [ ] Confirm camera model and interface
- [ ] Verify PSU voltage/current requirements
- [ ] Identify suppliers inside Germany/EU
- [ ] Place all orders and track shipping
- [ ] Create organizing folder for datasheets

PHASE 2 (Weeks 3–4) — Beginner Foundations

Checklist:

- [ ] Install Teensy/Arduino IDE + toolchain
- [ ] Blink LED on Teensy (GPIO test)
- [ ] Generate PWM on a pin
- [ ] Read Stepper driver datasheet
- [ ] Learn basic PID concept (P, I, D roles)
- [ ] Run simple PID simulation (optional MATLAB/Python)
- [ ] Document learning progress briefly

PHASE 3 (Weeks 5–6) — Initial Hardware Testing

Checklist:

- [ ] Wire Teensy to one stepper driver
- [ ] Connect driver to stepper motor
- [ ] Verify PSU output voltage
- [ ] Test step pulses + direction signals

- [x] Move motor a fixed number of steps
- [ ] Log behaviour (speed, noise, vibration)
- [ ] Create safety notes (max current values)

PHASE 4 (Weeks 7–8) — Single-Actuator Control

Checklist:

- [ ] Implement simple PID for position control
- [ ] Tune P term to get basic response
- [ ] Add I+D terms once stable
- [ ] Mount simple test platform
- [ ] Investigate camera model options (fps/latency)
- [ ] Take notes on actuator responsiveness

PHASE 5 (Weeks 9–10) — Mechanical Platform Design

Checklist:

- [x] Install Fusion 360 or SolidWorks
- [x] Model arm & motor mounts
- [x] Model platform geometry
- [x] Export CAD to STL/DXF if needed
- [x] Review torque requirements & arm lengths
- [x] Validate assembly fits together

PHASE 6 (Weeks 11–12) — Manufacturing & Assembly

Checklist:

- [x] 3D print/CNC all mechanical parts
- [ ] Sand/finish fits if required
- [ ] Assemble frame + platform
- [ ] Install motors + mounts
- [ ] Wire power, ground, signal neatly

- [ ] Add cable management (zip ties, sleeves)

PHASE 7 (Weeks 13–14) — Multi-Actuator Bring-Up

Checklist:

- [ ] Connect 2–4 stepper drivers to Teensy
- [ ] Move motors in coordinated pattern (sin wave/sequence)
- [ ] Test PSU load under motion
- [ ] Log actuator temperatures (optional)
- [ ] Check for mechanical interference
- [ ] Adjust step rates for smoothness

PHASE 8 (Weeks 15–16) — Camera Integration

Checklist:

- [ ] Test camera feed in Python (OpenCV)
- [ ] Measure real FPS & resolution
- [ ] Implement ball detection (color or circle detection)
- [ ] Display ball position overlay (visual debug)
- [ ] Log (x, y) ball coordinates
- [ ] Document lighting requirements

End of Feb

PHASE 9 (Weeks 17–18) — Vision + Control Integration

Checklist:

- [ ] Map camera pixels → platform coordinates
- [ ] Send ball coordinates to Teensy (USB/Serial)
- [ ] Implement closed-loop control logic
- [ ] Adjust delays & sampling rates
- [ ] Validate feedback with slow bounces
- [ ] Record short test videos

PHASE 10 (Weeks 19–20) — Full Closed-Loop Control

Checklist:

- [ ] Tune PID for multi-axis control
- [ ] Achieve continuous bounce
- [ ] Test different ball drop heights
- [ ] Measure latency end-to-end
- [ ] Video record stable operation

PHASE 11 (Weeks 21–22) — Experimental Evaluation

Checklist:

- [ ] Define performance metrics (e.g., stability, tracking error)
- [ ] Run repeated trials
- [ ] Save raw data logs
- [ ] Plot ball trajectory vs platform motion
- [ ] Summarize results in a lab notebook

PHASE 12 (Weeks 23–24) — Analysis & Improvements

Checklist:

- [ ] Identify control instability causes
- [ ] Improve mechanical stiffness if needed
- [ ] Improve PID or filtering (e.g., Kalman optional)
- [ ] Re-run experiments post improvements
- [ ] Capture final demo videos

PHASE 13 (Weeks 25–26) — Documentation Part 1

Checklist:

June - July

- [ ] Write Introduction + Problem Statement
- [ ] Write Background + Related Work
- [ ] Write Hardware & Mechanical Design

- [ ] Create system block diagrams
- [ ] Insert CAD screenshots + BOM

PHASE 14 (Weeks 27–28) — Documentation Part 2

Checklist:

- [ ] Write Control & Vision Methods
- [ ] Write Experiments + Results
- [ ] Write Discussion + Limitations + Future Work
- [ ] Format references + appendices

PHASE 15 (Weeks 29–30) — Revision & Submission

Checklist:

- [ ] Proofread entire thesis
- [ ] Check formatting rules (university style)
- [ ] Generate PDF + backups
- [ ] Prepare defense slides
- [ ] Submit thesis
- [ ] Demo prototype (optional)

📚 WEEKLY STUDY RESOURCE LIST (Beginner-Friendly)

Below are structured resources aligned to phases:

Weeks 1–2: BOM & Hardware

Study:

- Basic component datasheets
- How to choose steppers + drivers
- Basic voltage/current calculations

Resources:

- YouTube: "Stepper motors explained"
- Article: "How to choose a power supply"
- Datasheets from OMC/StepperOnline

Weeks 3–4: Microcontroller + PID Basics

Study:

- Teensy/Arduino programming
- PWM + Serial debugging
- PID basics

Resources:

- "PJRC Teensy getting started"
- Book: *PID Without a PhD* (Simple)
- YouTube: "PID control intro for beginners"

Weeks 5–6: Stepper Control

Study:

- Step/dir motor driving
- Microstepping
- Current limiting

Resources:

- "How stepper drivers work" videos
- OMC StepperOnline driver guides
- Arduino Stepper library examples

Weeks 7–8: Single Axis Control + Camera Basics

Study:

- PID tuning strategies
- Camera exposure & FPS concepts

Resources:

- MATLAB PID tuning demo (optional)
- YouTube: “Camera shutter, exposure, FPS basics”

Weeks 9–10: CAD Modelling

Study:

- Basic 3D modelling
- Joint mounting + tolerances

Resources:

- Fusion 360 “Designing Enclosures”
- YouTube: “Parametric modeling basics”

Weeks 11–12: Mechanical Assembly

Study:

- CNC/3D printing tolerances
- Cable management
- Power wiring safety

Resources:

- 3D printer calibration guides
- EEVblog power safety basics

Weeks 13–14: Multi Actuator Control

Study:

- Coordinated motion control
- Power distribution concepts

Resources:

- YouTube: “Coordinated motion control basics”
- Teensy Forum multi-axis projects

Weeks 15–16: Computer Vision (OpenCV)

Study:

- Color tracking
- Shape detection (Hough Circles)

Resources:

- OpenCV-Python Docs (ball tracking examples)
- YouTube: “OpenCV detect ball tutorial”

Weeks 17–18: Vision + Control Integration

Study:

- Serial communication PC → MCU
- Coordinate transformation

Resources:

- Tutorials on Serial/USB communication
- OpenCV + microcontroller integration blogs

Weeks 19–20: Real-Time Tuning

Study:

- Latency measurement
- Control loop frequency

Resources:

- YouTube: “Real-time control basics”
- Embedded timing tutorials

Weeks 21–22: Experimental Design

Study:

- How to collect data
- How to analyze response curves

Resources:

- SciPy/NumPy plotting tutorials
- MATLAB alternative: Python Matplotlib

Weeks 23–24: System Optimization

Study:

- Signal filtering (Simple low-pass)

- Mechanical damping concepts

Resources:

- YouTube: “Low-pass filter basics”
- Blogs on mechanical vibration reduction

Weeks 25–30: Documentation

Study:

- Thesis formatting
- Graph creation (Python/MATLAB)
- Technical writing clarity

Resources:

- “How to write a scientific thesis”
- Overleaf or Word templates
- IEEE citation basics